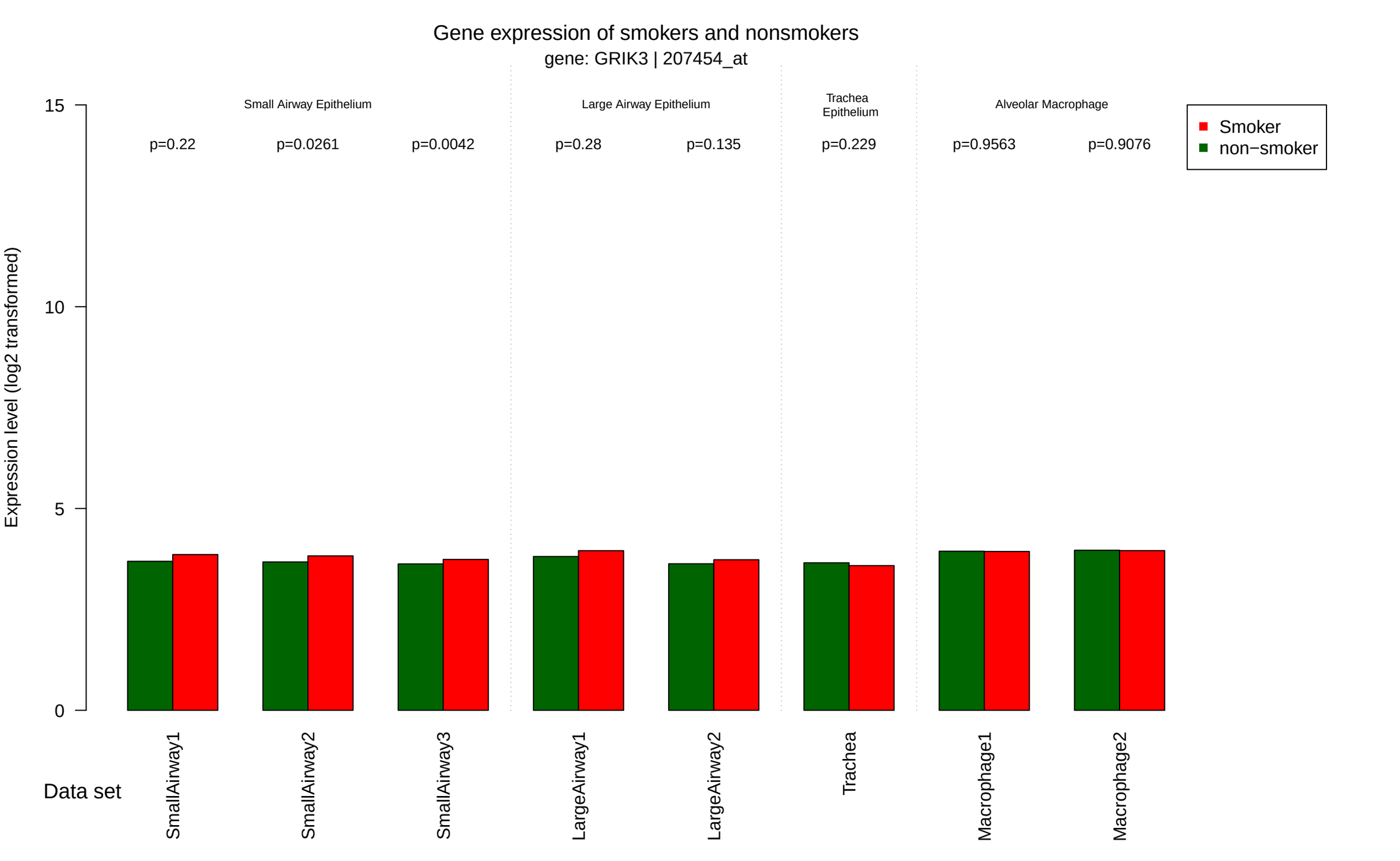
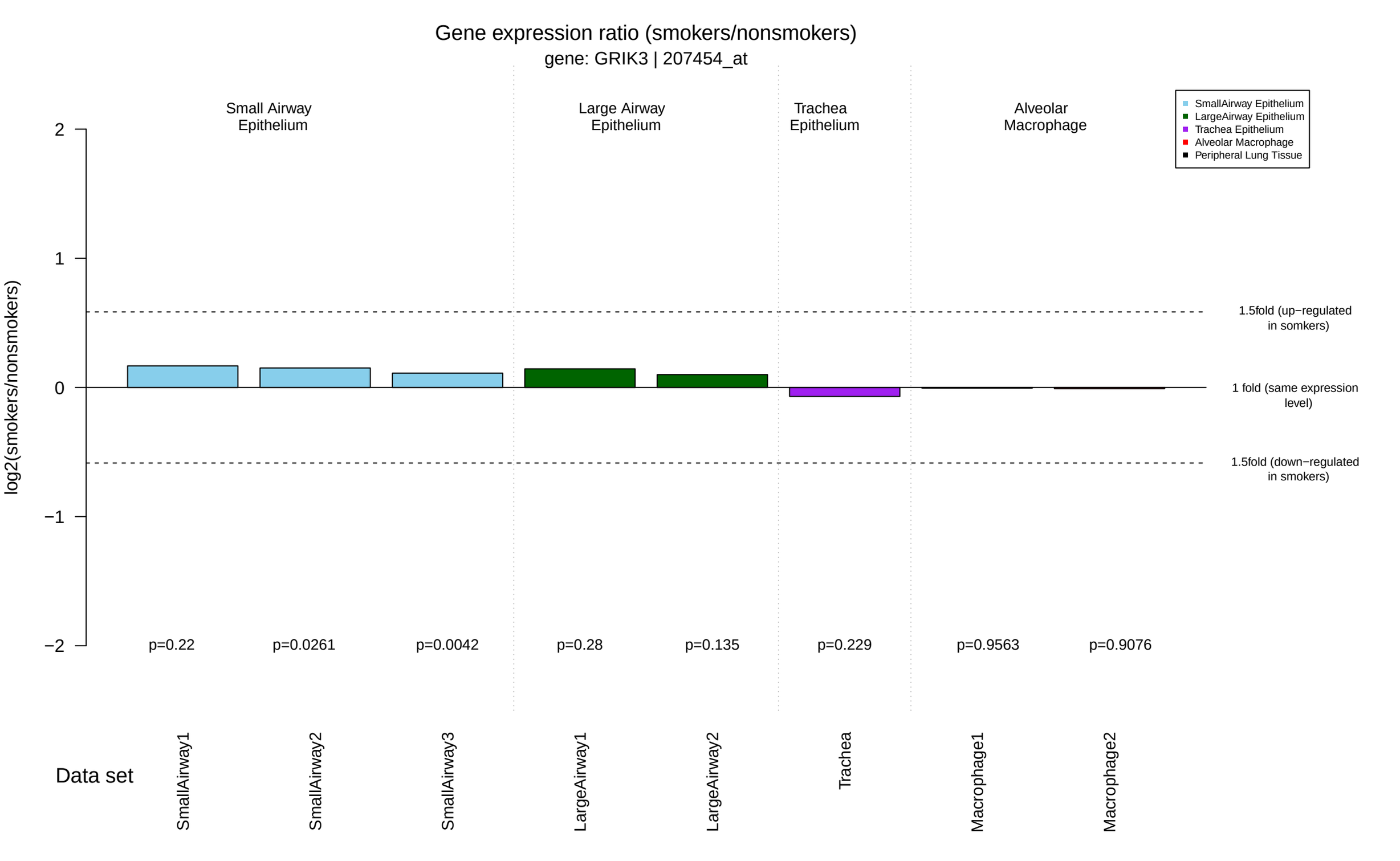
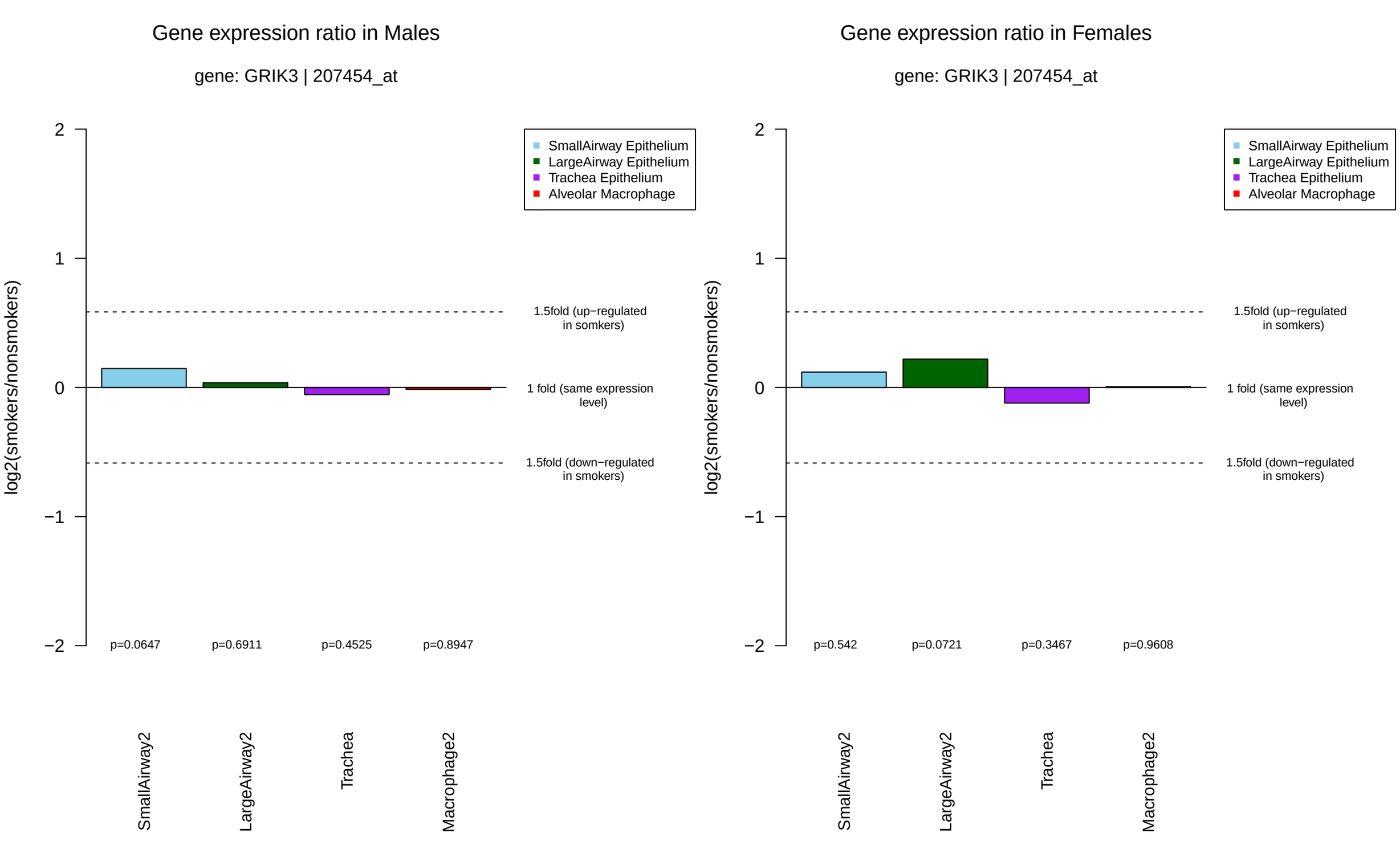
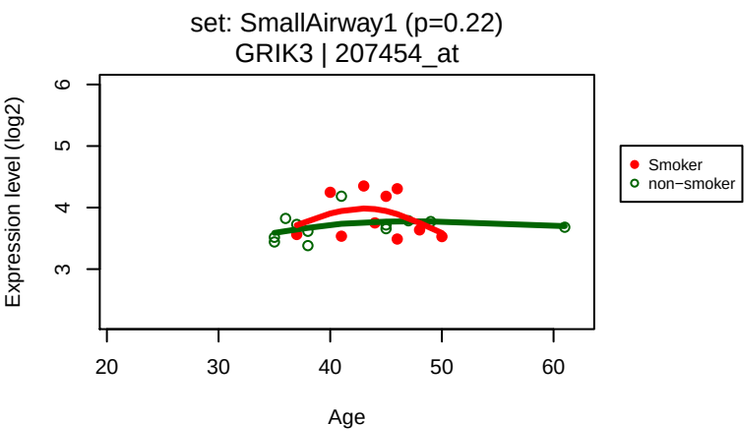
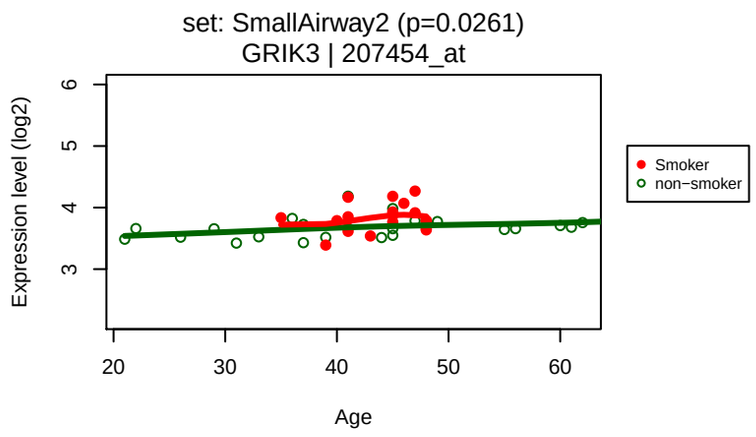
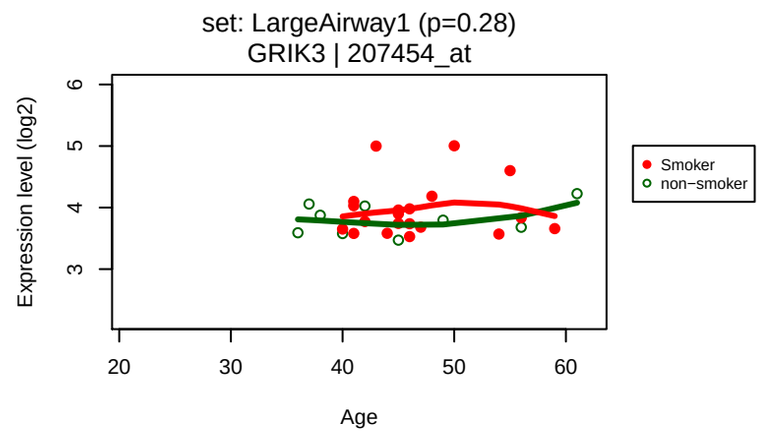

Gene expression of smokers and nonsmokers
gene: GRIK3 | 207454_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.22
p=0.0261
p=0.0042
p=0.28
p=0.135
p=0.229
p=0.9563
p=0.9076
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

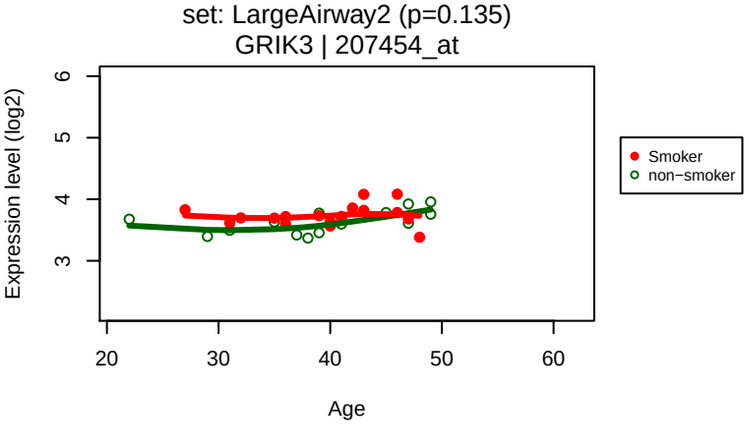

Gene expression ratio (smokers/nonsmokers)
gene: GRIK3 | 207454_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.22
p=0.0261
p=0.0042
p=0.28
p=0.135
p=0.229
p=0.9563
p=0.9076
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

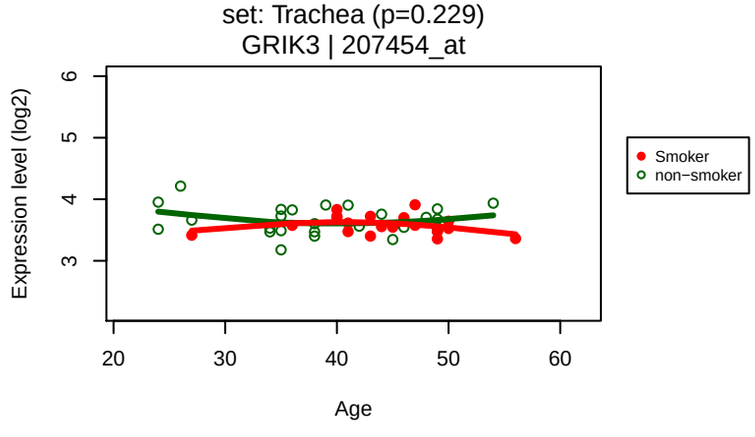

Gene expression ratio in Males
gene: GRIK3 | 207454_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0647
p=0.6911
p=0.4525
p=0.8947
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRIK3 | 207454_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.542
p=0.0721
p=0.3467
p=0.9608
SmallAirway2
LargeAirway2
Trachea
Macrophage2

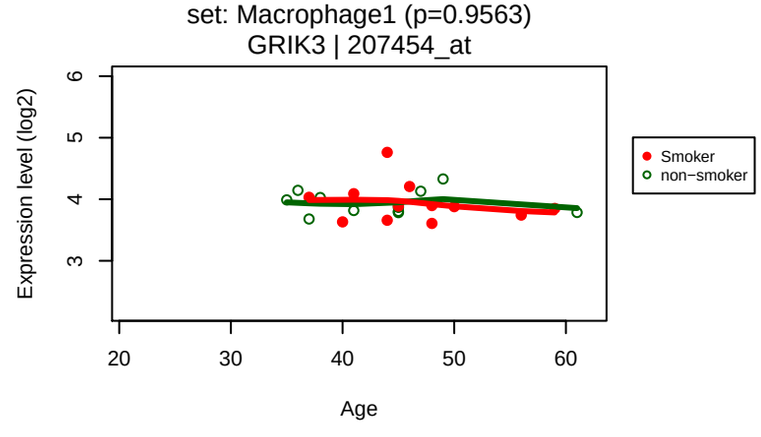

set: SmallAirway1 (p=0.22)
GRIK3 | 207454_at
Expression level (log2)
Age
Smoker
non−smoker

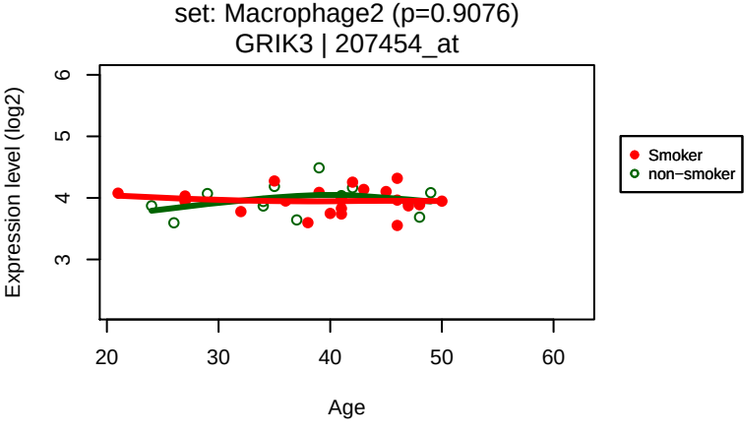

set: SmallAirway2 (p=0.0261)
GRIK3 | 207454_at
Expression level (log2)
Age
Smoker
non−smoker

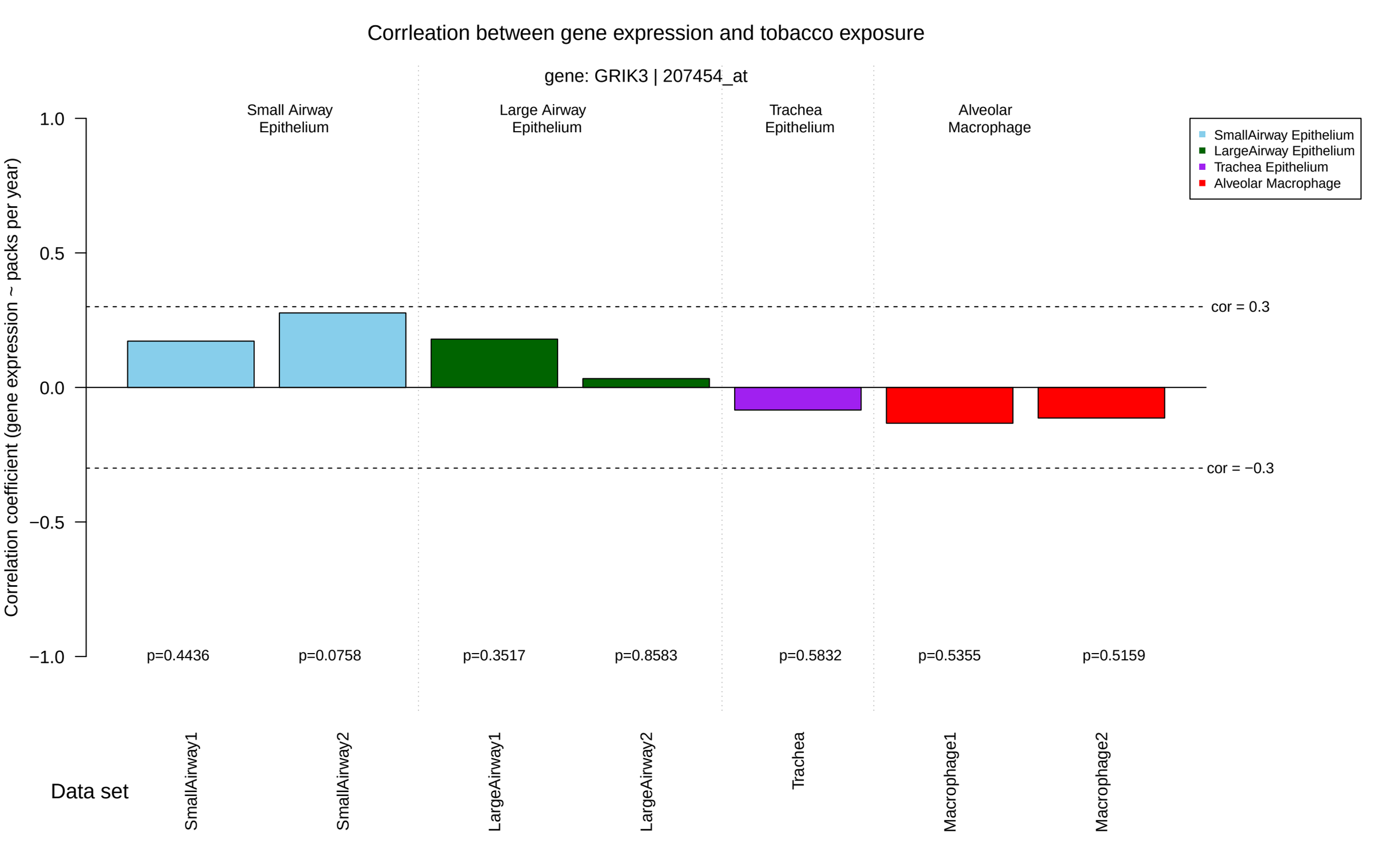

set: LargeAirway1 (p=0.28)
GRIK3 | 207454_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.135)
GRIK3 | 207454_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.229)
GRIK3 | 207454_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9563)
GRIK3 | 207454_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9076)
GRIK3 | 207454_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRIK3 | 207454_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4436
p=0.0758
p=0.3517
p=0.8583
p=0.5832
p=0.5355
p=0.5159
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2